

Manufacturing Downtime Analysis - Project Planning

Executive Summary

This document outlines the comprehensive project plan for the Manufacturing Downtime Analysis dashboard developed using Power BI. The project aims to provide real-time insights into production line performance, downtime factors, and operational efficiency metrics.

Project Duration: October 2025

Primary Tool: Microsoft Power BI Desktop

Data Source: Manufacturing_Line_Productivity.xlsx

1. Project Objectives

Primary Goals

- **Downtime Reduction:** Identify and analyze the top factors contributing to manufacturing downtime
- **Performance Monitoring:** Track Net Runtime, Total Downtime, and variance metrics across production lines
- **Data-Driven Decisions:** Enable management to make informed decisions based on real-time production data
- **Operational Visibility:** Provide clear visibility into operator and product performance

Key Performance Indicators (KPIs)

- Net Runtime Minutes
- Total Downtime Minutes
- Variance vs Minimum Runtime
- Downtime Percentage
- Top 5 Downtime Factors

2. Project Scope

In-Scope Deliverables

1. Overview Dashboard Page

- Three KPI cards showing Net Runtime, Downtime, and Variance
- Fourth KPI card for Downtime %
- Interactive slicers for Product, Operator, and Date filtering
- Clustered column chart showing Top 5 Downtime Factors
- Line chart displaying downtime trends over time
- Matrix visual showing Product × Operator performance breakdown
- Logo and branding elements

2. Data Model

- Three lookup tables: Products LU, Downtime factors LU, Line productivity LU
- One fact table: Line downtime LU
- Star schema relationships with one-to-many cardinality
- Date hierarchy for time-based analysis

3. DAX Measures

- Net Runtime Minutes
- Total Runtime Minutes
- Total Downtime Minutes
- Variance vs Min
- Downtime %

Out-of-Scope

- Batch-level detailed analysis (planned for Phase 2)
- Real-time data integration
- Mobile-optimized version
- Advanced predictive analytics

3. Project Timeline

Phase 1: Data Preparation (Week 1)

- Import Excel data source
- Create lookup tables (Products, Downtime Factors, Line Productivity)
- Establish star schema relationships
- Validate data integrity

Phase 2: Data Modeling (Week 1)

- Create calculated columns
- Build DAX measures
- Implement Date hierarchy
- Test measure accuracy

Phase 3: Dashboard Design (Week 2)

- Design Overview page layout
- Create KPI cards
- Build interactive slicers
- Develop Top 5 Downtime Factors chart
- Add trend line chart
- Create Product × Operator matrix

Phase 4: Formatting & Testing (Week 2)

- Apply consistent styling and color scheme
- Add logo and branding
- Test interactivity and filters
- Validate calculations
- Optimize performance

Phase 5: Documentation & Delivery (Week 3)

- Complete project documentation
- Prepare stakeholder presentation
- User training materials
- Final delivery and handoff

4. Resource Requirements

Human Resources

- **Business Analyst:** 1 FTE (Requirements gathering, stakeholder coordination)
- **Power BI Developer:** 1 FTE (Dashboard development, DAX calculations)
- **Data Engineer:** 0.5 FTE (Data preparation, quality checks)
- **Project Manager:** 0.25 FTE (Coordination, timeline management)

Technical Resources

- Microsoft Power BI Desktop (Latest version)
- Microsoft Excel (Data source management)
- Access to Manufacturing_Line_Productivity.xlsx
- Image editing tool for logo creation

Infrastructure

- Power BI Service (for future deployment)
- Network access to data sources
- Backup and version control system

5. Risk Management

Identified Risks

Risk	Impact	Probability	Mitigation Strategy
Data quality issues	High	Medium	Implement data validation rules and quality checks
Stakeholder requirement changes	Medium	High	Use iterative development with frequent reviews
Performance issues with large datasets	Medium	Low	Optimize DAX measures and use aggregations
User adoption challenges	Medium	Medium	Provide comprehensive training and documentation
Timeline delays	Low	Medium	Build buffer time into schedule

Contingency Plans

- **Data Issues:** Fallback to sample data for development, parallel data quality improvement
- **Technical Challenges:** Engage Microsoft support, consult Power BI community
- **Resource Constraints:** Prioritize core features, defer advanced features to Phase 2

6. Success Criteria

Quantitative Metrics

- Dashboard load time < 5 seconds
- 100% accuracy in DAX calculations vs manual Excel validation
- Support for filtering across 6+ products and 4+ operators
- Handle date range spanning 5+ days

Qualitative Metrics

- Positive stakeholder feedback on usability
- Management adoption for weekly production reviews
- Operator engagement with performance data
- Reduction in time spent on manual reporting

7. Communication Plan

Stakeholder Meetings

- **Weekly Status Updates:** Every Friday with project sponsors
- **Bi-weekly Reviews:** Demo sessions with end users
- **Daily Standups:** Development team sync (15 minutes)

Reporting

- Weekly progress reports
- Risk and issue logs
- Change request documentation
- Final project completion report

Escalation Path

1. Project Manager
2. Department Head
3. IT Director
4. Executive Sponsor

8. Quality Assurance

Testing Strategy

- **Unit Testing:** Validate each DAX measure independently
- **Integration Testing:** Verify filter interactions and cross-filtering
- **User Acceptance Testing:** End-user validation of functionality
- **Performance Testing:** Load testing with maximum expected data volume

Acceptance Criteria

- All KPIs calculate correctly
- Filters apply consistently across all visuals
- Top N filtering works as expected
- Date hierarchy enables drill-down
- Dashboard renders properly in 16:9 format
- All visuals have appropriate titles and labels

9. Training & Support

User Training

- **Admin Training:** 2-hour session covering data refresh and maintenance
- **End User Training:** 1-hour session on navigation and filtering
- **Train-the-Trainer:** Identify power users for peer support

Documentation Deliverables

- User guide with screenshots
- Administrator manual
- DAX measure documentation
- Troubleshooting guide

Post-Launch Support

- 30-day hyper-care period
- Dedicated support email
- Monthly check-ins for first quarter

10. Future Enhancements (Phase 2)

Planned Features

- Batch Detail page for granular analysis
- Predictive analytics for downtime forecasting
- Mobile-optimized views
- Real-time data integration
- KPI targets and variance alerts
- Drill-through to root cause analysis

Long-term Vision

- Integration with MES (Manufacturing Execution System)
- Automated report distribution
- AI-powered recommendations
- Cross-factory benchmarking dashboard

Conclusion

This project plan provides a comprehensive roadmap for delivering a high-quality Manufacturing Downtime Analysis dashboard. By following this structured approach, we ensure alignment with business objectives, efficient resource utilization, and successful delivery of a tool that will drive operational excellence in the manufacturing environment.